Problem Setup:

- **Data Points**: $(x, y) = (1, 2), (2, 3), (3, 5)$
- **Model**: $y = \theta_0 + \theta_1 x$
- **Loss Function**: Mean Squared Error (MSE) = $\frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$, where $m = 3$.
- **Learning Rate (α)**: 0.1
- **Initial Parameters**: $\theta_0 = 0, \theta_1 = 0$

Gradient Descent Update Rules:

1. $\theta_0 := \theta_0 - \alpha \cdot \frac{\partial J}{\partial \theta_0}$
2. $\theta_1 := \theta_1 - \alpha \cdot \frac{\partial J}{\partial \theta_1}$

Where:

- $\frac{\partial J}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$
- $\frac{\partial J}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$

Iteration Details:

Iteration	θ_0	θ_1	$Loss\ (J)$	$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$	$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$	Updated θ_0	Updated θ_1
0	0	0	12.67	-3.33	-8.67	0.33	0.867
1	0.33	0.867	4.22	-1.78	-4.44	0.511	1.311
2	0.511	1.311	1.41	-0.96	-2.38	0.607	1.549
3	0.607	1.549	0.47	-0.52	-1.28	0.659	1.677

Explanation of Each Column:

1. **Iteration**: The current iteration number.
2. θ_0: The current value of the intercept.
3. θ_1: The current value of the slope.
4. **Loss (J)**: The mean squared error for the current parameters.
5. $\frac{\partial J}{\partial \theta_0}$: The partial derivative of the loss with respect to θ_0.
6. $\frac{\partial J}{\partial \theta_1}$: The partial derivative of the loss with respect to θ_1.
7. **Updated** θ_0: The new value of θ_0 after the update.
8. **Updated** θ_1: The new value of θ_1 after the update.

Detailed Calculations for Iteration 0:

- **Predictions**: $h_\theta(x) = \theta_0 + \theta_1 x$
  - For $x = 1$: $0 + 0 \cdot 1 = 0$
  - For $x = 2$: $0 + 0 \cdot 2 = 0$
  - For $x = 3$: $0 + 0 \cdot 3 = 0$
- **Errors**: $h_\theta(x) - y$
  - For $x = 1$: $0 - 2 = -2$
  - For $x = 2$: $0 - 3 = -3$
  - For $x = 3$: $0 - 5 = -5$
- **Loss (J)**: $\frac{1}{2 \cdot 3}[(-2)^2 + (-3)^2 + (-5)^2] = \frac{1}{6}[4 + 9 + 25] = \frac{38}{6} \approx 6.33$
- **Gradients**:
  - $\frac{\partial J}{\partial \theta_0} = \frac{1}{3}[(-2) + (-3) + (-5)] = \frac{-10}{3} \approx -3.33$
  - $\frac{\partial J}{\partial \theta_1} = \frac{1}{3}[(-2 \cdot 1) + (-3 \cdot 2) + (-5 \cdot 3)] = \frac{1}{3}[-2 - 6 - 15] = \frac{-23}{3} \approx -7.67$
- **Updated Parameters**:
  - $\theta_0 := 0 - 0.1 \cdot (-3.33) = 0.333$
  - $\theta_1 := 0 - 0.1 \cdot (-7.67) = 0.767$

Iteration 1:

- **Predictions**: $h_\theta(x) = 0.333 + 0.767x$
  - For $x = 1$: $0.333 + 0.767 \cdot 1 = 1.1$
  - For $x = 2$: $0.333 + 0.767 \cdot 2 = 1.867$
  - For $x = 3$: $0.333 + 0.767 \cdot 3 = 2.634$

- **Errors**: $h_\theta(x) - y$
  - For $x = 1$: $1.1 - 2 = -0.9$
  - For $x = 2$: $1.867 - 3 = -1.133$
  - For $x = 3$: $2.634 - 5 = -2.366$
- **Loss (J)**: $\frac{1}{2 \cdot 3}[(-0.9)^2 + (-1.133)^2 + (-2.366)^2] = \frac{1}{6}[0.81 + 1.284 + 5.6] \approx 1.28$
- **Gradients**:
  - $\frac{\partial J}{\partial \theta_0} = \frac{1}{3}[(-0.9) + (-1.133) + (-2.366)] = \frac{-4.4}{3} \approx -1.467$
  - $\frac{\partial J}{\partial \theta_1} = \frac{1}{3}[(-0.9 \cdot 1) + (-1.133 \cdot 2) + (-2.366 \cdot 3)] = \frac{1}{3}[-0.9 - 2.266 - 7.098] = \frac{-10.264}{3} \approx -3.421$
- **Updated Parameters**:
  - $\theta_0 := 0.333 - 0.1 \cdot (-1.467) = 0.333 + 0.1467 = 0.4797$
  - $\theta_1 := 0.767 - 0.1 \cdot (-3.421) = 0.767 + 0.3421 = 1.1091$

Iteration 2:

- **Predictions**: $h_\theta(x) = 0.4797 + 1.1091x$
  - For $x = 1$: $0.4797 + 1.1091 \cdot 1 = 1.5888$
  - For $x = 2$: $0.4797 + 1.1091 \cdot 2 = 2.6979$
  - For $x = 3$: $0.4797 + 1.1091 \cdot 3 = 3.807$
- **Errors**: $h_\theta(x) - y$
  - For $x = 1$: $1.5888 - 2 = -0.4112$
  - For $x = 2$: $2.6979 - 3 = -0.3021$
  - For $x = 3$: $3.807 - 5 = -1.193$
- **Loss (J)**: $\frac{1}{2 \cdot 3}[(-0.4112)^2 + (-0.3021)^2 + (-1.193)^2] = \frac{1}{6}[0.169 + 0.091 + 1.423] \approx 0.28$

- **Gradients**:
  - $\frac{\partial J}{\partial \theta_0} = \frac{1}{3}[(-0.4112) + (-0.3021) + (-1.193)] = \frac{-1.9063}{3} \approx -0.6354$
  - $\frac{\partial J}{\partial \theta_1} = \frac{1}{3}[(-0.4112 \cdot 1) + (-0.3021 \cdot 2) + (-1.193 \cdot 3)] = \frac{1}{3}[-0.4112 - 0.6042 - 3.579] = \frac{-4.5944}{3} \approx -1.5315$
- **Updated Parameters**:
  - $\theta_0 := 0.4797 - 0.1 \cdot (-0.6354) = 0.4797 + 0.06354 = 0.54324$
  - $\theta_1 := 1.1091 - 0.1 \cdot (-1.5315) = 1.1091 + 0.15315 = 1.26225$

Iteration 3:

- **Predictions**: $h_\theta(x) = 0.54324 + 1.26225x$
  - For $x = 1$: $0.54324 + 1.26225 \cdot 1 = 1.80549$
  - For $x = 2$: $0.54324 + 1.26225 \cdot 2 = 3.06774$
  - For $x = 3$: $0.54324 + 1.26225 \cdot 3 = 4.32999$
- **Errors**: $h_\theta(x) - y$
  - For $x = 1$: $1.80549 - 2 = -0.19451$
  - For $x = 2$: $3.06774 - 3 = 0.06774$
  - For $x = 3$: $4.32999 - 5 = -0.67001$
- **Loss (J)**: $\frac{1}{2 \cdot 3}[(-0.19451)^2 + (0.06774)^2 + (-0.67001)^2] = \frac{1}{6}[0.0378 + 0.0046 + 0.4489] \approx 0.082$
- **Gradients**:
  - $\frac{\partial J}{\partial \theta_0} = \frac{1}{3}[(-0.19451) + (0.06774) + (-0.67001)] = \frac{-0.79678}{3} \approx -0.2656$
  - $\frac{\partial J}{\partial \theta_1} = \frac{1}{3}[(-0.19451 \cdot 1) + (0.06774 \cdot 2) + (-0.67001 \cdot 3)] = \frac{1}{3}[-0.19451 + 0.13548 - 2.01003] = \frac{-2.06906}{3} \approx -0.6897$
- **Updated Parameters**:
  - $\theta_0 := 0.54324 - 0.1 \cdot (-0.2656) = 0.54324 + 0.02656 = 0.5698$
  - $\theta_1 := 1.26225 - 0.1 \cdot (-0.6897) = 1.26225 + 0.06897 = 1.33122$

Final Parameters after 4 Iterations:

- $\theta_0 \approx 0.5698$
- $\theta_1 \approx 1.33122$

This table and the detailed calculations should help your students understand how gradient descent works in the context of linear regression.